Local Information-Enhanced Graph-Transformer for Hyperspectral Image Change Detection With Limited Training Samples

Wenqian Dong, *Member, IEEE*, Yufei Yang, *Student Member, IEEE*, Jiahui Qu, *Member, IEEE*, Song Xiao, and Yunsong Li, *Member, IEEE*

***Abstract*— Hyperspectral image change detection (HSI-CD) is a challenging task that focuses on identifying the differences between multitemporal HSIs. The recent advancement of convolutional neural network (CNN) has made great progress on HSIs-CD. However, due to the limited receptive field, most CNN-based CD models trained with sufficient labeled samples cannot flexibly model the global information that is essential for distinguishing complex objects, thereby achieving relatively low performance. In this article, we propose a dual-branch local information-enhanced graph-transformer (D-LIEG) CD network to fully exploit the local–global spectral–spatial features of the multitemporal HSIs with limited training samples for change recognition. Specifically, the proposed network is composed of a cascaded of LIEG blocks, which jointly extracts local–global features by learning local information representation to enhance the information of graph-transformer. A novel graph-transformer is developed to model global spectral–spatial correlation between graph nodes, enabling the spectral information preservation of HSIs and accurate CD of areas with various sizes. Extensive experiments have proved that our method achieves significant performance improvement than other state-of-the-art methods on four commonly used HSI datasets.**

***Index Terms*— Change detection (CD), graph-transformer, hyperspectral image (HSI), limited training samples.**

I. Introduction

CHANGE detection (CD) is a significant technique for identifying timely changes in nature scene by the use of multitemporal images over the same area. With the increasing availability of remote sensing satellite images, CD has been developed and widely used in massive remote sensing research activities, including natural disasters assessment [1], land-cover transitions [2], and water resource management [3]. As one of many remote sensing data, hyperspectral image (HSI) with hundreds of continuous spectral bands from the ultraviolet to mid-infrared wavelengths could demonstrate multiple spectral characteristics of land covers [4], [5], [6] and has become an effective tool for land-cover CD.

Numerous classic HSI-CD algorithms have been proposed and can be basically grouped in four categories, i.e., image algebra, transformation-based methods, classification-based methods, and other advanced algorithms [7], [8].

1) The image algebra methods calculate image difference between multitemporal data for acquiring CD results. The classical change vector analysis (CVA) [9] detects the changed regions by analyzing the Euclidian distance between the spectral information of the multitemporal images. Some variants of the CVA method are proposed in recent years to obtain more difference image, such as compressed CVA (C^2VA) [10] and sequential spectral CVA (S^2CVA) [11]. A novel framework is further proposed to combine the distance-based (i.e., image difference) and similarity-based (i.e., SAM) methods, which improves the performance of the existing CD methods [12]. However, these methods fail to consider the contextual information, which is sensitive to noise and misregistration errors.
2) Transformation-based methods, e.g., principal component analysis (PCA) [13], [14] and iteratively reweighed multivariate alteration detection (IR-MAD) [15], distinguish changes from unchanges by transforming HSI into a specific feature space. A synthetically fused image-based IR-MAD CD method further minimized the false alarm rate caused by registration error to increase the accuracy of CD [16]. Although resisting the noise interference, the transformation-based methods ignore the continuous spectral information of HSI and similarity between adjacent pixels.
3) Classification-based methods can be divided into postclassification and direct classification methods. The postclassification methods use a specific classifier, such as support vector machine (SVM) [17] and fuzzy c-means (FCM) [18], to discriminate two HSIs

Manuscript received 1 March 2023; revised 7 April 2023; accepted 19 April 2023. Date of publication 24 April 2023; date of current version 10 May 2023. This work was supported in part by the China Postdoctoral Science Special Foundation under Grant 2022T150508, in part by the Fundamental Research Funds for the Central Universities under Grant ZYTS23174, in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 095920221320, in part by the National Natural Science Foundation of China under Grant 62101414 and Grant 62201423, in part by the Youth Innovation Team of Shaanxi Universities, in part by the China Postdoctoral Science Foundation under Grant 2021M702546 and Grant 2021M702548, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2020A1515110856. *(Corresponding author: Jiahui Qu.)*

Wenqian Dong, Yufei Yang, Jiahui Qu, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: wqdong@xidian.edu.cn; yfyang_2021@stu.xidian.edu.cn; jhqu@xidian.edu.cn; yslixd@163.com).

Song Xiao is with the Department of Electronic and Communication Engineering, Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also with the School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: xiaosong@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3269892

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

separately, and extract the changed areas by comparing the classification results of multitemporal images. The direct classification methods treat the multitemporal images as a whole and detect changed areas with a classifier. The CD performance of classification-based methods depends heavily on the selection of classifier and the prior knowledge for training the classifier.

4) Other advanced models, such as texton forest [19], attribute profiles [20], and low rank-based [21], [22], [23] methods, have also been proven to be promising in CD.

The abovementioned methods are dependent on the hand-crafted shallow features, which are not powerful enough to represent the key information of the HSIs. Recently, deep learning-based approaches have obtained remarkable performance in many computer vision tasks [24], [25]. Lin et al. [26] introduced a bilinear convolutional neural network (BCNN) for multispectral image CD (MSI-CD), which trained the network with two symmetric CNNs. A general end-to-end 2-D CNN (GETNET) was developed in [27], which produces a mixed affinity matrix and inputs it into the 2-D CNN to detect the changes of HSIs. A three-direction spectral–spatial convolution neural network (TDSSC) [28] was designed to use 1-D and 2-D convolution to process high-dimensional change tensors from different directions, which fully learns the discriminant feature representations to improve the accuracy of HSI-CD. Dong et al. [29] hierarchically enhanced the correlation difference information at the subpixel level to detect subtle changes between bitemporal HSIs. Lyu et al. [30] proposed a CD approach named REFEREE, which relies on the long short-term memory (LSTM) to learn a stable changing rule. Mou et al. [31] proposed a recurrent CNN (ReCNN) for MSI-CD, which learns the joint spatial–spectral–temporal feature in an end-to-end way. Recently, some studies introduced graph convolutional network (GCN) to the field of HSI processing [32] for its powerful ability to model long-range spatial relations in the HSIs. Qu et al. [33] introduced GCN to HSI-CD to fully extract and effectively amplify the difference features. Wang et al. [34] proposed a novel module based on graph attention network (GAT) to fully exploit the temporal–spatial joint correlations of multitemporal HSIs.

Although the abovementioned methods have been proven effective in the task of HSI-CD, there are still some significant problems. Specifically, CNN-based methods that are severely constrained by the size of the convolution kernel only have a limited receptive field, which makes it difficult to model a global context that is essential for distinguishing objects in complex scenes. In addition, due to the high dimension of the sequence-based data structure of HSI in spectral domain, both CNN- and GCN-based methods have difficulty in capturing long-range sequential dependence between information of spectral bands, which is insufficient to exploit the abundant spectral information of HSIs.

Transformer, which uses self-attention to effectively model global dependency within sequential data, has shown good performance in many image processing tasks. However, the existing transformer-based methods take a series of patches with regular size as input, which fails to preserve the spectral information of HSI and learn discriminate features to deal with target objects with various shapes. The difficulty in the acquirement of labeled samples of HSIs also leads to the performance degradation of transformer. To address this issue, we propose a local information-enhanced graph-transformer (LIEG) CD network for HSIs, in which a dual-branch architecture composed of LIEG blocks is used to effectively model the global spectral–spatial dependency while maintaining local information. Since all pixels in each node are considered to belong to the same class, the label of the node is obtained even if only the label of one pixel is available. The proposed LIEG network taking the graph nodes as input can reduce the number of samples, making it easier to obtain sufficient labeled nodes to satisfy the training of the proposed network. Our main contributions can be summarized as follows.

1) We design a novel graph-transformer to fully consider the global spectral–spatial dependency between graph nodes, which well preserves spectral information of HSIs and can better deal with irregular changed areas.
2) The LIEG block composed of the graph-transformer with global representation capability and the convolution layer with local acquisition capability is innovatively proposed, which effectively extracts local and global features simultaneously by using the locality of convolution to enhance the information of the graph-transformer.
3) The dual-branch structure composed of multilayer LIEG blocks is designed to fully extract complementary local and global features of multitemporal HSIs to realize the discrimination of changed and unchanged objects with limited training samples.

The rest of this article is organized as follows. The related works are reviewed in Section II. The proposed method is described in detail in Section III. Section IV provides experimental results on four datasets. Finally, Section V concludes this article.

II. Related Work

Recently, studies have given much attention to transformer to alleviate the drawbacks of CNN and GCN, and many transformer-based methods have been proposed for image processing tasks. The transformer model was originally proposed for natural language processing (NLP) tasks [35]. Due to its strong ability in learning long-range sequential dependence of input data, many efforts have been made to apply the transformer and its variants to visual tasks and have achieved remarkable success, such as detection transformer (DETR) [36], vision transformer (ViT) [37], and data-efficient image transformers (DEIT) [38]. In addition, transformers have also attracted widespread attention and achieved remarkable success on other computer vision tasks of natural images, such as image generation [39] and image enhancement [40].

Inspired by the aforementioned works, transformer is introduced to the field of remote sensing image processing, such as light detection and ranging (LiDAR) and HSI classification, high-resolution optical remote sensing image CD, and HSI target detection. For example, HSI-bidirectional encoder representations from transformers (BERT) [41] adopted a variant of the transformer (BERT) to HSI classification task, which captures global dependence between pixels regardless of their spatial distance. Xue et al. [42] further applied transformer to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

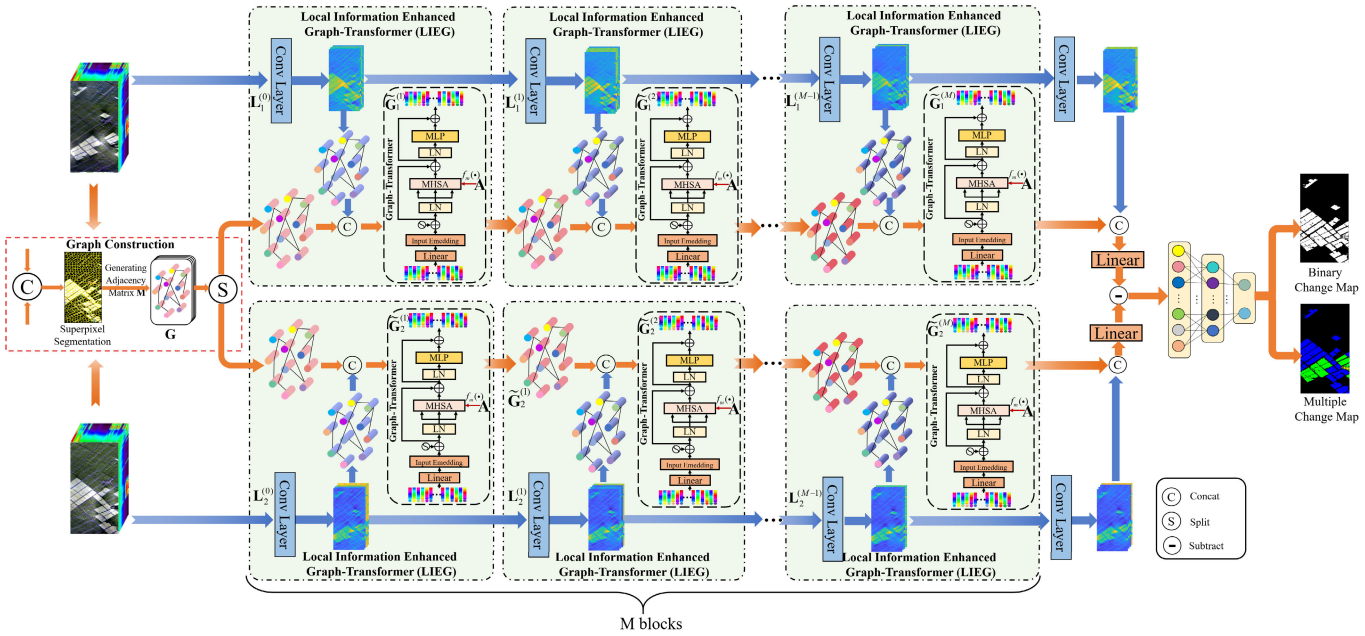


Fig. 1. Overall architecture of the proposed method, which converts the multitemporal HSIs into graph according to the spectral–spatial similarity and constructs a dual-branch CD network with a cascade of LIEG blocks to fully extract the local–global spectral–spatial features of the multitemporal HSIs with limited training samples.

multimodality data collaborative classification, i.e., hyperspectral and LiDAR data classification, which modifies the original ViT to extract the spatial features of HSI and LiDAR. For high-resolution image CD that can be regarded as binary classification task of multitemporal images, Chen et al. [43] utilized transformer encoder to model long-range contexts of the multitemporal images in the compact token-based space time. Wang et al. [44] further designed a multiscale transformer to learn contextual feature representations at different scales in multitemporal high-resolution images. In order to simultaneously consider the spatial, spectral, and temporal information of multitemporal HSIs, Wang et al. [45] proposed spectral–spatial transformer and temporal transformer for HSI-CD.

Although the transformer with the advantage of modeling global information has achieved good performance in many remote sensing image processing tasks, there are still many challenges in the application of HSI-CD. Since the methods designed for high-resolution remote sensing images receive a series of regular patches that flatten all the pixels into a vector, directly transferring them to HSI-CD will lead to the loss of spectral information. To preserve the spectral information of HSIs, the existing methods for HSI take the spectral vector of all pixels as input, which ignores the context information and leads to high computational cost. To tackle these problems, we propose a graph-transformer to model global spectral–spatial dependency of all graph nodes, in which each node considers the spectral–spatial similarity and the irregular changed areas can be better detected. The dual-branch LIEG (D-LIEG) network for HSI-CD further promotes the representation capability of graph-transformer by injecting the local information, thus modeling the global-local spectral–spatial features of HSIs.

III. Proposed Method

In this section, the notation is described first; then, the framework is analyzed, and the details of the proposed HSI-CD network are further explained.

A. Notation

Before describing the proposed method, we introduce some notations used in this article. The multitemporal HSIs obtained at time t_1 and t_2 are, respectively, represented by matrixes as $\mathbf{T}_1 \in \mathbb{R}^{HW\times C}$ and $\mathbf{T}_2 \in \mathbb{R}^{HW\times C}$, in which H and W, respectively, denote the height and width of the image, and C is the number of bands of the image. Each column of the image matrix corresponds to a spectral band, which contains $H \times W$ pixels arranged in order. Given the input HSIs $\mathbf{T}_1 \in \mathbb{R}^{HW\times C}$ and $\mathbf{T}_2 \in \mathbb{R}^{HW\times C}$, the goal of HSI-CD is to obtain a change map $\widehat{\mathbf{Y}} \in \mathbb{R}^{H\times W}$ reflecting whether the object at the corresponding position in the two multitemporal HSIs has changed.

B. Framework Overview

In this article, we propose a new transformer that takes graph nodes as input sequence called graph-transformer, which efficiently models the global context of the image. Since local information is also crucial for recognizing objects in complex scenes, we use convolution to model local information and inject it into the graph-transformer to extract local–global features. The local–global features of multitemporal HSIs are fully explored by a dual-branch CD network to facilitate the discrimination of changed and unchanged objects. As shown in Fig. 1, the proposed architecture is mainly composed of three parts, including graph-transformer, LIEG block, and D-LIEG CD network. Specifically, the two multitemporal HSIs are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

cascaded, and then, superpixel segmentation is performed, which can effectively obtain the distribution of graph nodes and the mapping relationship between the graph and the original image. Then, the graph-transformer is proposed to model the relationship between the sequence of graph nodes. In order to make up for the lack of local information, the LIEG block is designed in which the local information obtained by convolution is injected into the graph nodes. In this way, the multilayer LIEG blocks are used to construct a dual-branch CD network to fully extract the local–global features of the multitemporal HSIs to determine whether the corresponding pixels between the multitemporal HSIs have changed or not.

C. Graph-Transformer

The transformer based on the self-attention mechanism can well model the relationship between the embedded sequences, which is suitable for processing high-dimensional HSIs with complex scenes. For HSIs data, taken the spectral vector corresponding to each pixel as an input sequence, transformer has global receptive field to capture richer global dependence between each pair of pixels, thus modeling the context of the whole image. However, it is obvious that considering the spectral vector of each pixel individually will result in a huge amount of calculation, which is unacceptable especially for large-sized images. To reduce the amount of calculation, Dosovitskiy et al. [37] split the whole image into a large number of regular patches and send them to the transformer for image recognition. However, taking a series of patches with regular size as input fails to preserve the spectral information of HSI and learn discriminate features to deal with target objects with various shapes. To solve the above problems, we innovatively propose a novel transformer named graph-transformer, in which the whole HSI is divided into a series of compact graph nodes as the input of the transformer. We segment the HSI according to the spectral–spatial similarity, and the average spectral vector value of the pixels contained in the segmented region is determined as the feature vector value of the corresponding graph node for spectral information preservation. In this way, not only the computational complexity can be greatly reduced, but also the spectral–spatial correlation between pixels can be fully considered.

To construct graph nodes, a region-based segmentation method named simple linear iterative clustering (SLIC) [46] is introduced, which iteratively grows local clusters gradually via a K-means algorithm until the iteration reaches the optimum to complete the segmentation operation. Pixels with high spectral–spatial similarity are usually divided into the same image regions (i.e., graph nodes), so that each graph node has a high probability to belong to the same land-cover type. The label of the node can be obtained even if only the label of one pixel is known, making the proposed network overcome the limitation of labeled samples for HSI-CD. Specifically, in order to ensure that the multitemporal HSIs have consistent graph node distribution, the two input HSIs $\mathbf{T}_1$ and $\mathbf{T}_2$ are cascaded along the channel dimension and then divided into a series of compact region. The dimension of the cascaded image is reduced to 3 to adapt the input of the SLIC algorithm. In this article, the multitemporal HSIs are divided into N graph nodes. The feature vector value of each node is obtained by taking the average of spectral vector of all pixels in the segmented region. Therefore, the cascaded data are converted to an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of nodes and edges, respectively. The weights of edge are defined by the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$. In many GCN-based methods, Gaussian kernel function is adopted to define the weight of each pair of nodes, which calculates the Euclidean distance to measure the similarity between two connected nodes. Considering the abundant spectral information contained in HSIs, we obtain the adjacency matrix $\mathbf{A}$ by calculating the spectral angle mapper (SAM) between each pair of nodes to evaluate the similarity of them. For each node, the values of $\mathbf{A}$ at the corresponding position of the first r nodes (the number of neighbor nodes r is set to 10 in this article) with the highest similarity are set to 1, and the rest are 0. In this way, the information of neighbor nodes with high probability of belonging to the same ground object type is integrated to update the current node, effectively avoiding the interference of information of different ground object types. The adjacency matrix at position (p, q) $\mathbf{A}_{pq}$ is calculated as follows:

$$\mathbf{A}_{pq} = \begin{cases} 1, & \text{if } \mathcal{R}\big(\text{SAM}(\mathbf{x}_p, \mathbf{x}_q)\big) \\ & \quad = \mathcal{R}\left(\arccos\left(\dfrac{\langle \mathbf{x}_p, \mathbf{x}_q \rangle}{\|\mathbf{x}_p\| \|\mathbf{x}_q\|}\right)\right) \le r \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathbf{x}_p$ and $\mathbf{x}_q$ represent the pth and qth superpixels, respectively, and $\mathcal{R}(\cdot)$ denotes the function of the rank of the SAM value between $\mathbf{x}_p$ and $\mathbf{x}_q$ among that between $\mathbf{x}_p$ and each superpixels. The original feature $\mathbf{G}$ of the graph can be expressed as follows:

$$\mathbf{G} = \mathbf{M}^{\mathrm{T}} \text{Concat}(\mathbf{T}_1, \mathbf{T}_2) \tag{2}$$

where $\mathbf{M} \in \mathbb{R}^{HW\times N}$ is the association matrix between segmentation result and original image and Concat$(\cdot)$ is the feature concatenation across channels. In this case, graph nodes instead of pixels or patches are regarded as the input of the graph-transformer, which greatly improves the efficiency of the algorithm and effectively ensures the spectral–spatial similarity of pixels in the graph node.

For HSI-CD tasks, the ability of graph-transformer to model context is particularly effective for distinguishing objects in complex scenes. As shown in Fig. 2, the graph-transformer usually receives a 1-D sequence of graph node as input. In practice, a set of input sequences that are simultaneously sent to the graph-transformer are packed into a matrix. In order to unify the dimensions of the input matrix to facilitate the training of the network, we perform input embedding operation on the feature vector of the input graph nodes, which transforms the feature vector into the embedding with dimension C_u by linear projection. Then, the position encoding is added to maintain position information. Given the original feature of multitemporal HSIs $\mathbf{G}_i \in \mathbb{R}^{N\times C}$, the obtained embedding matrix $\mathbf{I}_i \in \mathbb{R}^{N\times C_u}$ is regarded as the input of the graph-transformer.

The graph-transformer is generally cascaded by n encoder layers (here, $n = 3$), where each encoder layer is mainly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

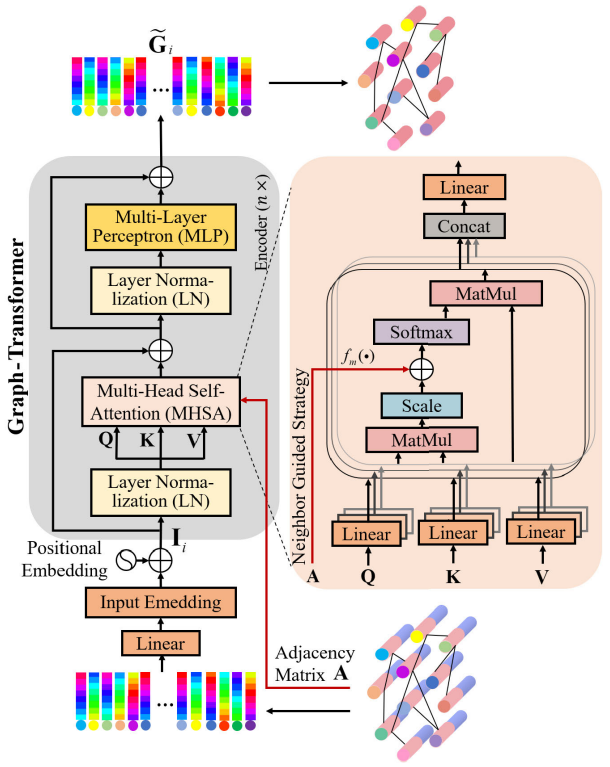


Fig. 2. Overall architecture of the proposed graph-transformer, which takes a series of compact graph nodes as the input and models the global spectral–spatial dependency between graph nodes to better deal with irregular changed areas.

composed of multihead self-attention (MHSA), multilayer perceptron (MLP), layer normalization (LN), and residual connection. The key to the self-attention mechanism is using a query and a set of key–value pairs to get an attention map to focus on the influence between pixels at different locations. In addition, we further apply the adjacency matrix $\mathbf{A}$ to the attention operation $\mathrm{Att}(\cdot)$ in the proposed graph-transformer, which aims to give higher weights to the nodes with high similarity. Therefore, the useful information is emphasized, and the interference of redundant information is suppressed. The neighbor-guided attention operation $\mathrm{Att}(\cdot)$ is defined as follows:

$$\mathrm{Att}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}} + f_m(\mathbf{A})\right)\mathbf{V} \tag{3}$$

where $\mathbf{Q}\in\mathbb{R}^{N\times d}$, $\mathbf{K}\in\mathbb{R}^{N\times d}$, and $\mathbf{V}\in\mathbb{R}^{N\times d}$ represent the query, key, and value, respectively, and $f_m(\cdot)$ denotes the function that converts 0 to $-\infty$ and 1 to 0 to match the property of softmax function. The function $f_m(\cdot)$ can be expressed as follows:

$$f_m(\mathbf{A}) = \begin{cases} 0, & \text{if } \mathbf{A}_{pq}=1 \\ -\infty, & \text{if } \mathbf{A}_{pq}=0. \end{cases} \tag{4}$$

Instead of simply executing a single attention function, MHSA adopts an MHSA mechanism, in which each head obtains discriminative contextual feature representations in parallel to improve the ability of feature learning. The embedded input $\mathbf{I}_i$ is subjected to LN that can effectively alleviate the internal covariate shift of the deep learning model during the training process and then fed to MHSA with s heads $(h_1, h_2, \ldots, h_s)$ to explore the context information, which is defined as follows:

$$\begin{aligned}\mathrm{MHSA}(\mathrm{LN}(\mathbf{I}_i)) &= \mathrm{Concat}[h_1,h_2,\ldots,h_s]\mathbf{W}^0, \quad \text{where} \\ h_j &= \mathrm{Att}\big(\mathbf{Q}_j,\mathbf{K}_j,\mathbf{V}_j\big) \\ &= \mathrm{Att}\big(\mathrm{LN}(\mathbf{I}_i)\mathbf{W}_j^q, \mathrm{LN}(\mathbf{I}_i)\mathbf{W}_j^k, \mathrm{LN}(\mathbf{I}_i)\mathbf{W}_j^v\big) \\ &= \mathrm{Att}\big(\mathrm{LN}\big(\big(\mathbf{B}\mathbf{M}^T\mathbf{T}_i\big)\mathbf{E}+\mathbf{P}\big)\mathbf{W}_j^q, \\ &\qquad \mathrm{LN}\big(\big(\mathbf{B}\mathbf{M}^T\mathbf{T}_i\big)\mathbf{E}+\mathbf{P}\big)\mathbf{W}_j^k, \\ &\qquad \mathrm{LN}\big(\big(\mathbf{B}\mathbf{M}^T\mathbf{T}_i\big)\mathbf{E}+\mathbf{P}\big)\mathbf{W}_j^v\big)\end{aligned} \tag{5}$$

where $i = 1$ and 2, the number of heads s is set to 8, $\mathbf{B}\in\mathbb{R}^{N\times N}$ is a diagonal matrix with $\mathbf{B}_{ii} = (1/\sum_j \mathbf{A}_{ij})$, $\mathbf{P}\in\mathbb{R}^{N\times C_u}$ denotes the position encoding, $\mathbf{E}\in\mathbb{R}^{C\times C_u}$, $\mathbf{W}_j^q \in \mathbb{R}^{C_u\times d}$, $\mathbf{W}_j^k\in\mathbb{R}^{C_u\times d}$, $\mathbf{W}_j^v\in\mathbb{R}^{C_u\times d}$, and $\mathbf{W}^0\in\mathbb{R}^{sd\times C_u}$ are the learned parameter matrices, $\mathrm{Concat}[\cdot]$ denotes the concatenate operation, and $\mathrm{LN}(\cdot)$ represents the LN layer. In addition, a residual connection is adopted to avoid the loss of feature information, and the obtained feature $\mathbf{I}_i'$ can be formulated as follows:

$$\mathbf{I}_i' = \mathrm{MHSA}(\mathrm{LN}(\mathbf{I}_i)) + \mathbf{I}_i. \tag{6}$$

The MLP composed of two linear transformation layers and a Gaussian error linear unit (GELU) activation function is then introduced to further transform the learning features of all heads. The results obtained by MHSA are sequentially sent to the LN and MLP for feature integration. The output of the graph-transformer $\widetilde{\mathbf{G}}_i$ can be given as follows:

$$\widetilde{\mathbf{G}}_i = \mathrm{MLP}(\mathrm{LN}(\mathbf{I}_i')) + \mathbf{I}_i' \tag{7}$$

where $\mathrm{MLP}(\cdot)$ denotes the function of MLP block.

In summary, the forward propagation process of the entire graph-transformer can be briefly described as follows:

$$\widetilde{\mathbf{G}}_i = f_{\mathrm{gt}}(\mathbf{I}_i, \mathbf{A}) \tag{8}$$

where $\widetilde{\mathbf{G}}_i \in \mathbb{R}^{N\times C_u}$ is the output of graph-transformer and $f_{\mathrm{gt}}(\cdot)$ denotes the function of graph-transformer.

D. LIEG Block

The graph-transformer that benefits from the self-attention mechanism can well model the long-term contextual relationship between sequences of graph nodes in parallel. However, the feature vector of each graph node is the average value of the spectral vector of the pixels contained in the node, which inevitably leads to partial loss of local information in the node. Convolution operation provides a very effective way to extract local information, which utilizes local receptive fields and shared weights to effectively capture the correlationship between adjacent pixels and model the local information of the image. Therefore, we inject the local information extracted by convolution into the graph-transformer to deal with the lack of local information. A new LIEG block is designed, which combines the advantages of convolution in locality and the advantages of graph-transformer in modeling global context to obtain the feature representations with local–global information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

The LIEG with a strong ability to extract local–global features can effectively identify objects in multitemporal HSI to obtain different features for CD. Specifically, for the lth LIEG block, the convolutional feature map of the $l-1$th block is fed into a convolutional layer to extract local features of the lth block, and the kernel size of the convolutional layer in all the blocks is 3×3 with stride 1, and the filter number of each convolutional layer is set to 256. Then, the extracted local features of the lth block are mapped into a feature matrix $\overline{\mathbf{G}}^{(l)}$ by left multiplying the transpose of the association matrix. The obtained feature matrix $\overline{\mathbf{G}}^{(l)}$ can be defined as follows:

$$\overline{\mathbf{G}}^{(l)} = \mathbf{M}^{\mathrm{T}}(\mathrm{Conv}(\mathbf{L}^{(l)})) \tag{9}$$

where $\mathbf{L}^{(l)}$ denotes the convolutional feature map of the $l-1$th block and $\mathrm{Conv}(\cdot)$ represents the local information extraction convolution operation. The local feature matrix of the lth block $\overline{\mathbf{G}}^{(l)}$ and the output feature matrix of the $l-1$th block $\widetilde{\mathbf{G}}^{(l)}$ are concatenated along the channel dimension and sent to a linear projection to obtain the input of the graph-transformer in the lth LIEG block $\mathbf{I}_{\mathrm{LE}}^{(l)} \in \mathbb{R}^{N \times C_u}$ as follows:

$$\mathbf{I}_{\mathrm{LE}}^{(l)} = f_{\mathrm{linear}}(\mathrm{Concat}[\widetilde{\mathbf{G}}^{(l)}, \overline{\mathbf{G}}^{(l)}]) \tag{10}$$

where f_{linear} is the linear projection function. As a result, the sequences of graph nodes enhanced with local information are sent to the graph-transformer to extract local–global feature. The forward model of LIEG can be simplified as follows:

$$\begin{cases} \widetilde{\mathbf{G}}^{(l+1)} = f_{\mathrm{gt}}\left(\mathbf{I}_{\mathrm{LE}}^{(l)}, \mathbf{A}\right) \\ \mathbf{L}^{(l+1)} = \mathrm{Conv}\left(\mathbf{L}^{(l)}\right) \end{cases} \tag{11}$$

where $\widetilde{\mathbf{G}}^{(l+1)}$ and $\mathbf{L}^{(l+1)}$ represent the output feature matrix and the convolutional feature map of the lth LIEG block, respectively.

E. D-LIEG: Dual-Branch LIEG CD Network

With excellent ability to extract local and global information, LIEG is especially suitable for identifying the feature representation of regions of interest in multitemporal HSI-CD tasks. Therefore, we propose a D-LIEG HSI-CD network embedded in multiple LIEG blocks. Specifically, the proposed network adopts a dual-branch structure to obtain sufficient features of multitemporal HSIs to distinguish changed and unchanged objects. Each branch is composed of multiple LIEG blocks to extract complementary local and global features. The propagation process of multilayer LIEG can be described as follows:

$$\left(\widetilde{\mathbf{G}}_i^{(m)}, \mathbf{L}_i^{(m)}\right) = f_{\mathrm{LIEG}}, \ldots, f_{\mathrm{LIEG}}\left(\widetilde{\mathbf{G}}_i^{(0)}, \mathbf{L}_i^{(0)}, \mathbf{A}\right) \tag{12}$$

where $i = 1$ and 2, $m \in [1, M]$ is the mth LIEG blocks on each branch, $\mathbf{G}_i^{(m)}$ and $\mathbf{L}_i^{(m)}$, respectively, represent the output feature matrix and the convolutional feature map of the mth LIEG block on the ith branch, with $\mathbf{G}_i^{(0)} = \mathbf{G}_i$, $\mathbf{L}_i^{(0)} = \mathbf{T}_i$, and $f_{\mathrm{LIEG}}(\cdot)$ is a simplified description of the forward propagation process of LIEG. In order to further preserve the local information, the convolution output of the last LIEG block is sent to a convolutional layer and concatenated with the output of the last block, which is defined as follows:

$$\mathbf{B}_i = f_{\mathrm{linear}}\left[\mathrm{Concat}\left(\widetilde{\mathbf{G}}_i^{(M)}, \mathbf{M}^{\mathrm{T}}\mathrm{Conv}\left(\mathbf{L}_i^{(M)}\right)\right)\right] \tag{13}$$

where $\mathbf{B}_i$ is the output of the ith branch.

The output of the two branches containing sufficient features is subtracted to obtain the difference features and then transformed back to the feature map by using the association matrix. The obtained feature map is sent to a classifier composed of two fully connected layers, a rectified linear unit (ReLU) activation function, and a softmax nonlinear function to further obtain the prediction results of CD as follows:

$$\widehat{\mathbf{Y}} = \mathrm{softmax}\left\{f_{\mathrm{fc2}}\left[f_{\mathrm{fc1}}(\mathbf{M}(\mathbf{B}_1 - \mathbf{B}_2))\right]\right\} \tag{14}$$

where f_{fc1} and f_{fc2} represent the fully connected layers and $\widehat{\mathbf{Y}}$ is the output prediction map.

During the network training process, the cross-entropy loss function is used to optimize the network parameters, which can be expressed as follows:

$$\mathrm{CE} = -\sum_{a=1}^{n} y_a \log \hat{y}_a \tag{15}$$

where n is the number of classes, y_a represents the label of reference image, and $\hat{y}_a$ is the probability of class a.

IV. Experiments

A. Experimental Settings

1) Datasets: In this article, four public datasets are used to test the validity of the proposed D-LIEG method. The first dataset is the Santa Barbara dataset[1] and was captured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The two multitemporal HSIs of the Santa Barbara dataset, which was captured in Santa Barbara area in 2013 and 2014, have 250×250 pixels and include 224 bands covering the wavelength range from 400 to 2500 nm, as shown in Fig. 3(a) and (b). The Bay Area dataset[1] is the second dataset, and the two multitemporal HSIs of this dataset were captured using AVIRIS sensors in Paterson in 2013 and 2015, as shown in Fig. 4(a) and (b). Each HSI consists of 250×250 pixels and includes 224 bands in a wavelength range of 400–2500 nm. The ground truth of the Santa Barbara and Bay Area datasets are, respectively, shown in Figs. 3(c) and 4(c), where yellow indicates changed regions, green indicates unchanged regions, and blue indicates unmarked regions. The third dataset is the Farmland dataset,[2] and the two multitemporal HSIs were taken from the farmland near Yancheng city, Jiangsu Province, China, on May 3, 2006 and April 23, 2007, as shown in Fig. 5(a) and (b). Each HSI is captured by the Earth Observing-1 (EO-1) Hyperion hyperspectral sensor, which consists of 450×140 pixels and 154 bands, covering the spectral range from 400 to 2500 nm. The ground truth of binary CD of the Farmland dataset is

[1]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[2]https://rslab.ut.ac.ir/data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

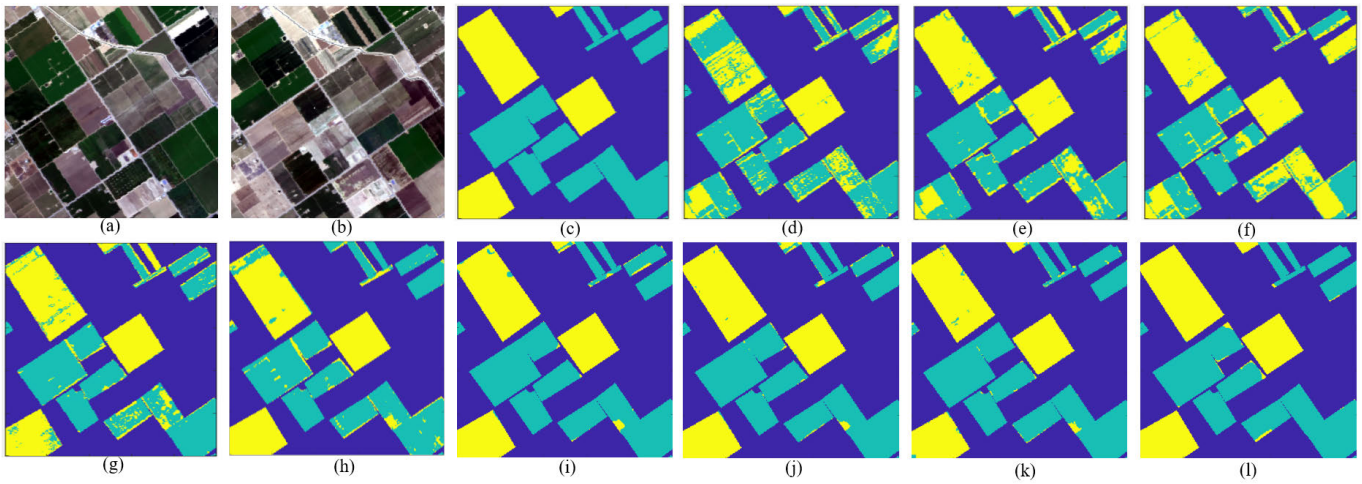


Fig. 3. Subjective visual comparison of binary CD result of each method for the Santa Barbara dataset. (a) and (b) Multitemporal HSIs of the Santa Barbara dataset. (c) Ground truth. (d) CVA. (e) PCA. (f) IR-MAD. (g) SVM. (h) ReCNN. (i) BCNNs. (j) BIT. (k) ML-EDAN. (l) Proposed.

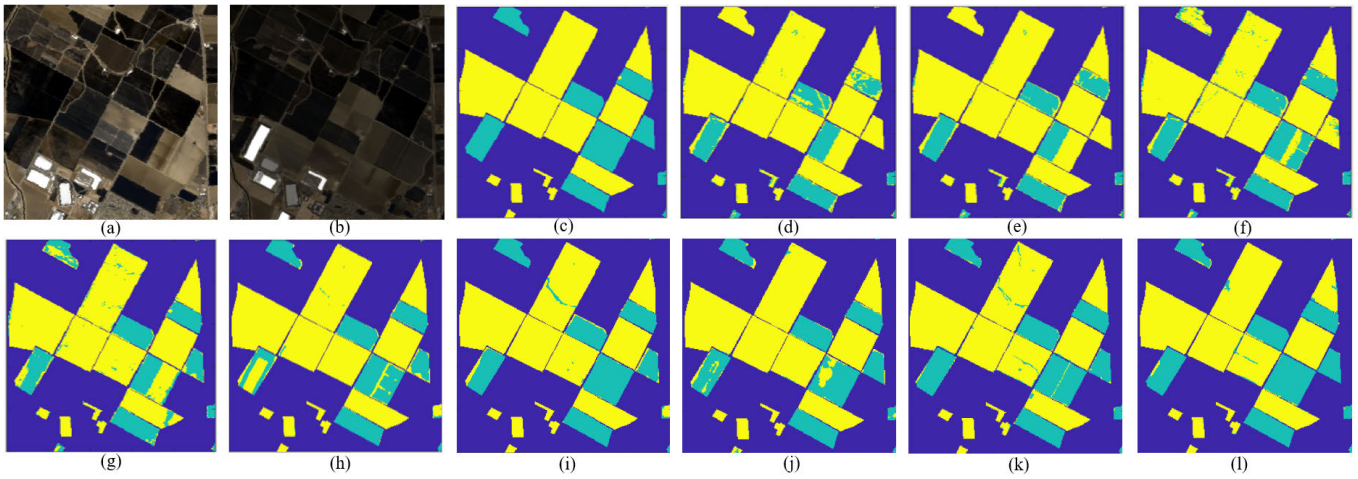


Fig. 4. Subjective visual comparison of binary CD result of each method for the Bay Area dataset. (a) and (b) Multitemporal HSIs of the Bay Area dataset. (c) Ground truth. (d) CVA. (e) PCA. (f) IR-MAD. (g) SVM. (h) ReCNN. (i) BCNNs. (j) BIT. (k) ML-EDAN. (l) Proposed.

shown in Fig. 5(c), and white represents changed regions and black represents unchanged regions. The last dataset is the Hermiston dataset,[3] which was acquired by the EO-1 Hyperion sensor on May 1, 2004 and May 8, 2007. This dataset covers an irrigated agricultural field in Hermiston City, Umatilla County, OR, USA. The spatial dimensions of HSIs are 225 × 180 pixels, and 159 spectral bands are used in the experiments. The multitemporal HSIs are shown in Fig. 6(a) and (b), and the corresponding ground-truth maps for binary CD is shown in Fig. 6(c). The ground truth maps of Farmland and Hermiston datasets for multiple CD are shown in Figs. 7(a) and 8(a), respectively.

2) Evaluation Metrics and Implementation Details: To quantitatively evaluate the performance of each HSI-CD methods, overall accuracy (OA) and kappa coefficient (KC) are calculated, where OA measures the overall accuracy of each method and KC measures the consistency of the predicted results with the ground-truth map. F1-score is also a widely used metric that calculates the harmonic mean of precision and recall. Specifically, precision reflects the false alarm rate, and recall reflects the miss alarm rate. We compare the CD performance of the proposed D-LIEG method and the compared methods with the limited training sample size of 0.5% and 1% in this article.

The proposed method is implemented on NVIDIA GTX 3090 GPU based on the PyTorch framework. For training, we utilize the Adam optimizer with a learning rate of $1e-5$ and finish learning procedure after 800 epochs. In the experiments, the numbers of superpixels of the Santa Barbara, Bay Area, Farmland, and Hermiston datasets are set to 300, 150, 100, and 70, respectively.

B. Parameter Setting

1) Value of r: To investigate the effect of the value of r, we test the performance of the proposed method with the value of r from 5 to 20 in steps of 5, and the performance of case that all pixels are taken into consideration is also investigated. The OA, KC, and F1-score of the results are

[3]https://github.com/SicongLiuRS/Hyperspectral-Change-Detection-Dataset-Irrigated-Agricultural-Area

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

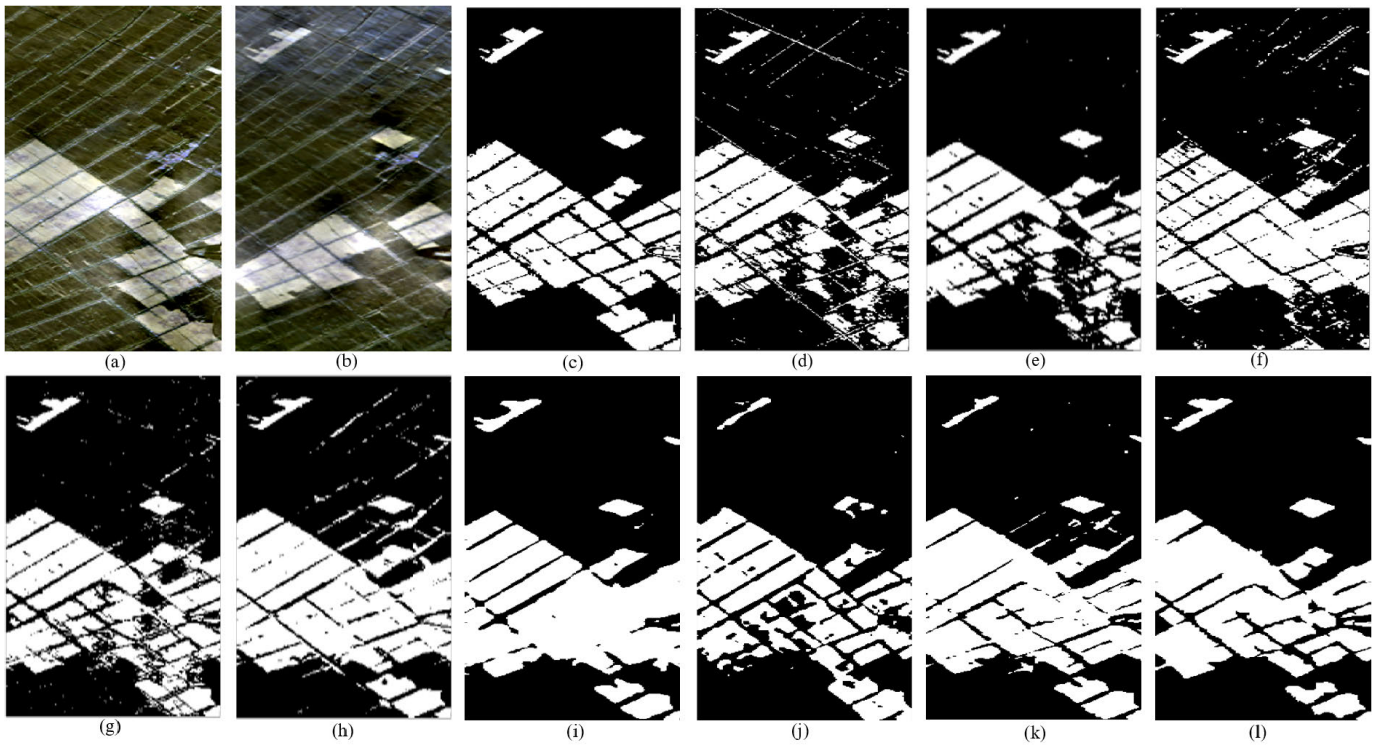


Fig. 5. Subjective visual comparison of each method for the Farmland dataset. (a) and (b) Multitemporal HSIs of the Farmland dataset. (c) Ground truth. (d) CVA. (e) PCA. (f) IR-MAD. (g) SVM. (h) ReCNN. (i) BCNNs. (j) BIT. (k) ML-EDAN. (l) Proposed.

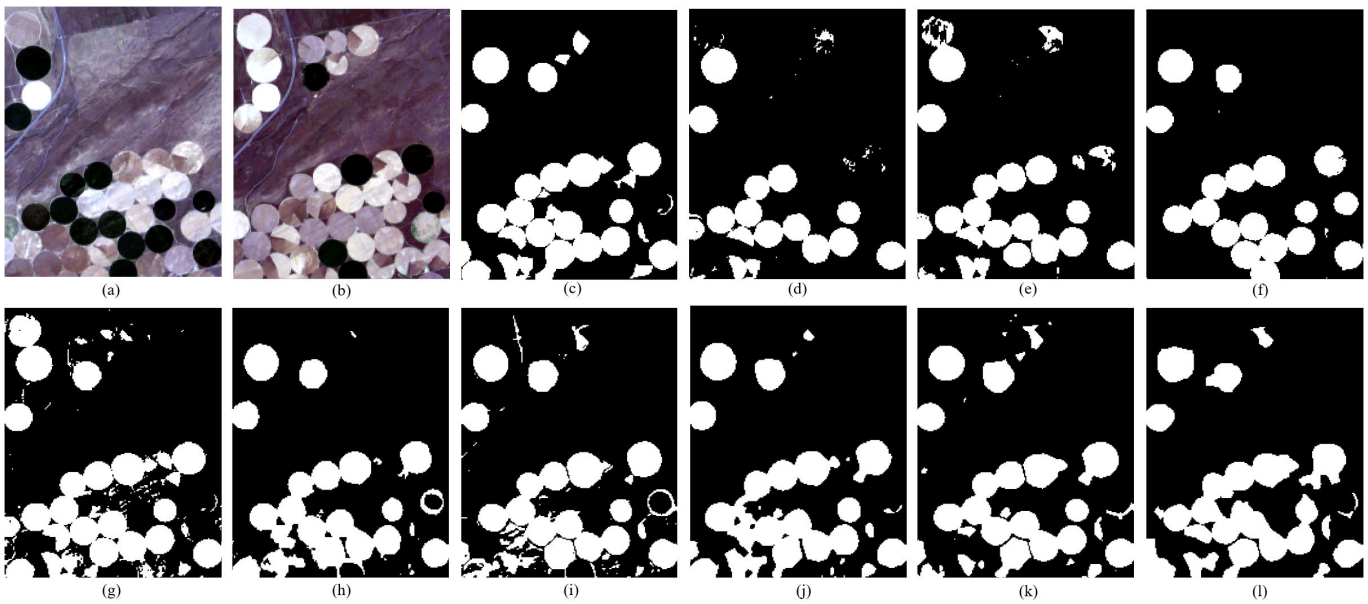


Fig. 6. Subjective visual comparison of binary CD result of binary CD result of each method for the Hermiston dataset. (a) and (b) Multitemporal HSIs of the Farmland dataset. (c) Ground truth. (d) CVA. (e) PCA. (f) IR-MAD. (g) SVM. (h) ReCNN. (i) BCNNs. (j) BIT. (k) ML-EDAN. (l) Proposed.

shown in Fig. 9. As can be seen, the performance of the model is unsatisfactory when all pixels are considered in the self-attention mechanism, indicating the effectiveness of neighbor-guided attention operation that only considers the neighbor nodes with the highest similarity. The model with $r = 10$ obtains better CD accuracy than the model with $r = 5$. When $r > 10$, the performance of the model degrades as the value of r increasing. This is because too large value of r would introduce interference information of nodes with different land-cover types. The results show that the most similar nodes are selected when $r = 10$, so the value of r is set to 10 in the proposed method.

2) Number of Superpixel: The proposed method adopts the SLIC algorithm to segment the whole image into superpixels. Since there exist differences in the size of HSI and the scale of ground objects, the optimal numbers of superpixel are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

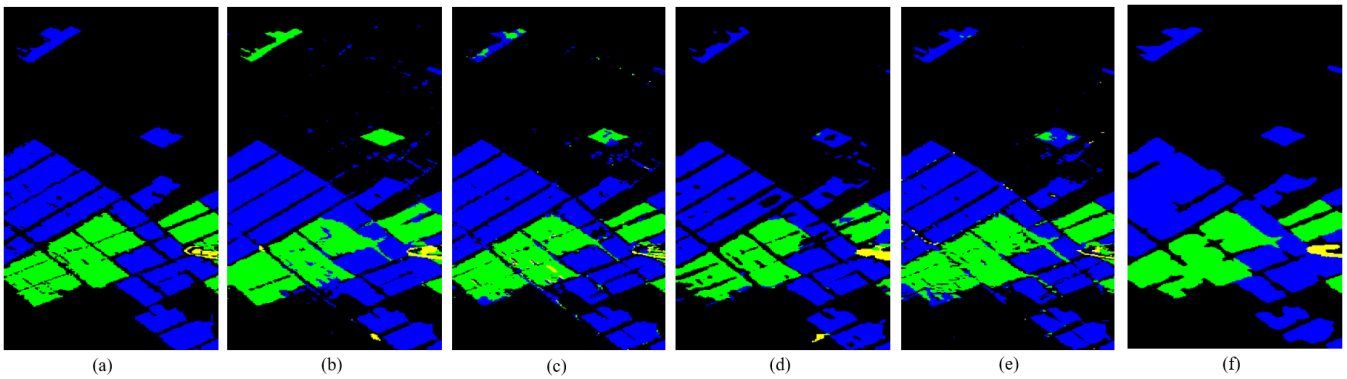


Fig. 7. Subjective visual comparison of multiple CD result of each deep learning-based method for the Farmland dataset. (a) Ground truth. (b) ReCNN. (c) BCNNs. (d) BIT. (e) ML-EDAN. (f) Proposed.

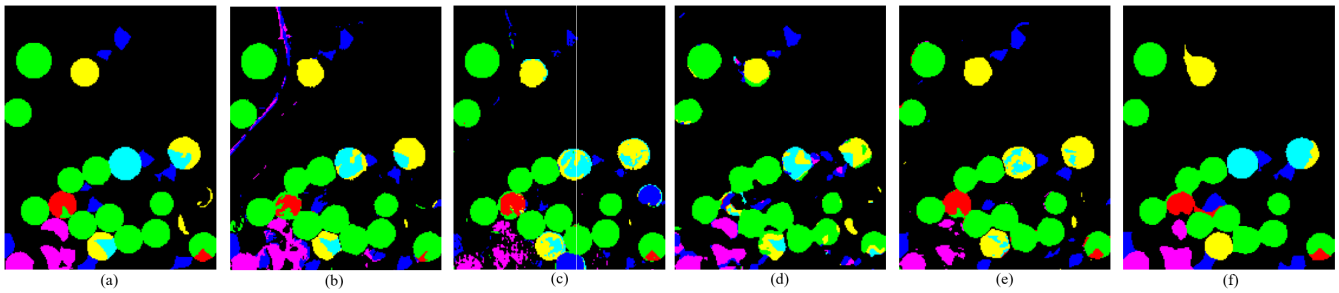


Fig. 8. Subjective visual comparison of multiple CD result of deep learning-based method for the Hermiston dataset. (a) Ground truth. (b) ReCNN. (c) BCNNs. (d) BIT. (e) ML-EDAN. (f) Proposed.

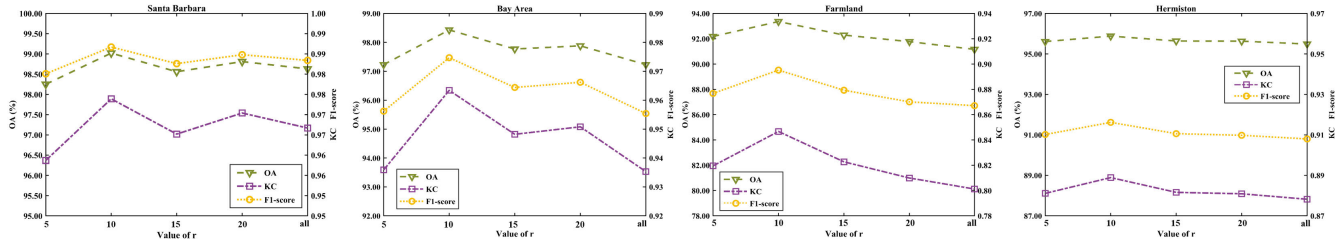


Fig. 9. Performance of the proposed method with different values of r on four datasets.

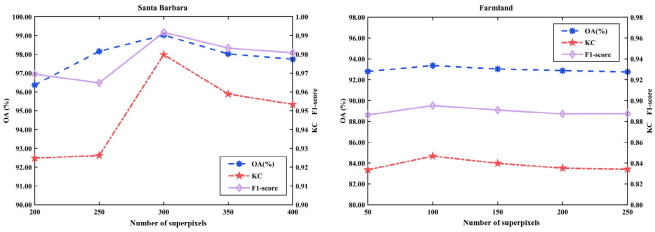


Fig. 10. Performance of the proposed method with different numbers of superpixels on Santa Barbara and Farmland datasets.

different on different datasets. To find the superpixel number that make the method achieve the best results, we conduct experiments on all the datasets. We set the step size of superpixel number as 50, and the experimental results of Santa Barbara and Farmland datasets are shown in Fig. 10. The performance of the method is satisfactory with different numbers of superpixel. According to the values of OA, KC, and F1-score, we choose the numbers of superpixel as 300, 150, 100, and 70 on the Santa Barbara, Bay Area, Farmland, and Hermiston datasets to make the network obtain the best results, respectively.

C. Experimental Results

1) Binary CD: To verify the effectiveness of the proposed D-LIEG method, we select eight widely used HSI-CD approaches, including CVA [9], PCA [13], IR-MAD [15], SVM [17], ReCNN [26], BCNNs [31], bitemporal image transformer (BIT) [43], and multilevel encoder–decoder attention network (ML-EDAN) [47]. Figs. 3–6 show the subjective results of each competing methods on four datasets with the limited training sample size of 1%. The CVA method has some noise points in the change map, since it measures the Euclidean distance between pixels and is sensitive to noise. PCA and IR-MAD improve performance by reducing redundant information in the spectrum, but still cannot effectively detect some boundary areas. The performance of the SVM method is unsatisfactory, and many changed areas cannot be effectively detected. ReCNN detects some changed pixels, but a large number of unchanged areas are erroneously detected as changed areas. BCNNs achieve better performance than ReCNN with less white noise points, but there still exist some missed detected areas. BIT fails to preserve the details of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OBJECTIVE RESULTS OF BINARY CD OF EACH METHOD ON FOUR DATASETS

Dataset		Santa Barbara		Bay Area		Farmland		Hermiston	
Training Sample size		0.5%	1%	0.5%	1%	0.5%	1%	0.5%	1%
CVA	OA(%)	72.46		85.63		87.49		89.10	
	KC	0.4256		0.6202		0.6998		0.6638	
	F1-score	0.7713		0.7060		0.7882		0.7277	
PCA	OA(%)	79.13		87.12		88.91		90.69	
	KC	0.5752		0.6635		0.7321		0.7297	
	F1-score	0.8135		0.7442		0.8041		0.7884	
IRMAD	OA(%)	70.86		90.53		89.74		91.73	
	KC	0.4282		0.7637		0.7632		0.7594	
	F1-score	0.7760		0.8216		0.8382		0.8115	
SVM	OA(%)	87.93	89.75	90.69	91.77	89.88	90.20	91.74	93.83
	KC	0.7524	0.7919	0.7684	0.7984	0.7640	0.7730	0.7905	0.8382
	F1-score	0.8868	0.9182	0.8259	0.8559	0.8343	0.8365	0.8464	0.8795
ReCNN	OA(%)	93.81	95.60	93.49	94.78	89.90	91.13	92.71	94.02
	KC	0.8520	0.8960	0.8650	0.8940	0.7970	0.8220	0.7942	0.8394
	F1-score	0.9441	0.9628	0.9008	0.9170	0.9213	0.8738	0.8412	0.8791
BCNNs	OA(%)	95.33	98.39	93.81	97.68	91.61	92.56	92.85	94.17
	KC	0.9034	0.9666	0.8469	0.9451	0.8117	0.8321	0.8018	0.8405
	F1-score	0.9605	0.9863	0.8891	0.9617	0.8743	0.8874	0.8485	0.8788
BIT	OA(%)	94.28	98.56	94.02	96.86	91.37	91.86	92.57	95.35
	KC	0.8827	0.9702	0.8618	0.9249	0.7891	0.8037	0.7955	0.8734
	F1-score	0.9508	0.9878	0.9056	0.9471	0.8487	0.8608	0.8443	0.9041
ML-EDAN	OA(%)	95.89	98.47	95.58	98.16	92.20	92.69	93.40	95.27
	KC	0.9154	0.9683	0.8955	0.9570	0.8267	0.8341	0.8064	0.8694
	F1-score	0.9649	0.9871	0.9272	0.9704	0.8852	0.8883	0.8475	0.9004
Proposed	OA(%)	**96.54**	**99.02**	**96.05**	**98.43**	**92.70**	**93.36**	**94.25**	**95.88**
	KC	**0.9286**	**0.9798**	**0.9087**	**0.9634**	**0.8316**	**0.8467**	**0.8375**	**0.8889**
	F1-score	**0.9706**	**0.9917**	**0.9376**	**0.9747**	**0.8859**	**0.8952**	**0.8746**	**0.9162**

HSIs, which leads to false detection on the boundary of the changed regions. ML-EDAN can better preserve the high-level information than BIT, obtaining higher CD accuracy. It can be clearly seen that the results obtained by the proposed D-LIEG method are closest to the ground truths. This mainly because compared with the competing methods, the proposed method could fully extract the global-local spectral–spatial features of HSIs, thereby better dealing with irregular changed areas.

The objective results of each method with the limited training sample size of 0.5% and 1% are shown in Table I. It can be observed that the traditional methods, such as CVA, PCA, and IR-MAD, provide low OA, KC, and F1-score values on all datasets. The results indicate that the traditional methods cannot fully exploit the deep features in the HSIs, resulting in serious false detection and missed detection phenomenon. Compared with the unsupervised methods, the accuracy of the supervised methods (including SVM, ReCNN, BCNNs, BIT, and ML-EDAN) is significantly higher. The accuracy of BIT is slightly lower than that of BCNNs. Benefiting from the hierarchical multilevel features, ML-EDAN performs better than other methods. From Table I, the OA, KC, and F1-score values obtained by the proposed D-LIEG are the highest on the four datasets with the training sample size of both 0.5% and 1%. The experimental results demonstrate that the proposed network outperforms the competing methods, indicating that the global context can be effectively extracted and the local information is well maintained.

TABLE II
OBJECTIVE RESULTS OF MULTIPLE CD OF DEEP LEARNING-BASED METHODS ON FARMLAND AND HERMISTON DATASETS

Dataset	Farmland				
Method	ReCNN	BCNNs	BIT	ML-DEAN	Proposed
Unchanged	0.9248	0.9503	0.9704	0.9421	0.9421
Change-1	0.8790	0.8820	0.8770	0.9036	0.9298
Change-2	0.8862	0.8619	0.7480	0.8805	0.8648
Change-3	0.5517	0.3724	0.8690	0.3241	0.6414
OA(%)	91.04	92.56	92.79	92.63	**93.90**
KC	0.8179	0.8441	0.8337	0.8468	**0.8562**
F1-score	0.7660	0.7525	0.6628	0.7609	**0.8111**
Dataset	Hermiston				
Method	ReCNN	BCNNs	BIT	ML-DEAN	Proposed
Unchanged	0.9465	0.9665	0.9851	0.9863	0.9814
Change-1	0.7242	0.3845	0.4971	0.7185	0.6861
Change-2	0.9765	0.9534	0.8783	0.9184	0.8992
Change-3	0.8573	0.7859	0.6725	0.8501	0.6574
Change-4	0.5678	0.7683	0.0292	0.9019	0.8706
Change-5	0.6741	0.6032	0.5385	0.3097	0.7561
Change-6	0.8259	0.5832	0.7708	0.5435	0.8404
OA(%)	92.75	92.32	92.16	93.78	**94.29**
KC	0.8294	0.8099	0.8007	0.8425	**0.8575**
F1-score	0.8030	0.7506	0.6972	0.8004	**0.8239**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

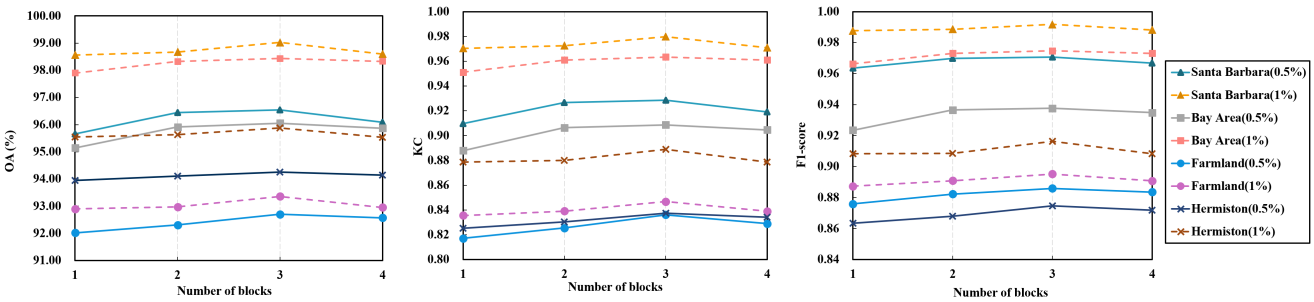


Fig. 11. Performance of the proposed method with different LIEG block numbers on four datasets.

2) Multiple CD: The multiple CD results for Farmland and Hermiston datasets are shown in Figs. 7 and 8. Since the performances of traditional CD methods are poor, only the results of deep learning-based methods with a training sample size of 1% are used for comparison. It can be observed that ReCNN and BCNNs produce noisy change maps with many misclassified unchanged areas, especially the boundaries of ground objects. BIT shows better performance and reduces much of the noise, but there are serious missed detection phenomena even in the smooth areas. Although there still exist some false detection areas, ML-DEAN shows relatively good performance in both datasets. Benefiting from the graph-transformer that models the global spectral–spatial dependency between graph nodes, the proposed method reduces the false detection and missed detection to a great extent, showing the best visual result.

The quantitative results on Farmland and Hermiston datasets are listed in Table II. Although BIT shows relatively good performance in OA and KC, the value of F1-score is low. The values of OA, KC, and F1-score of ML-DEAN are higher than other competing methods. The proposed method obtains the best result in terms of all metrics on both Farmland and Hermiston datasets, which indicates a high detection accuracy and a balance between false detection and missed detection.

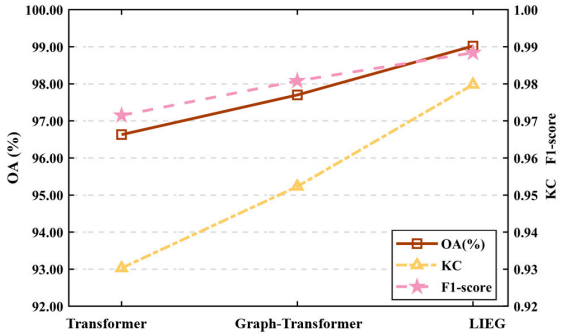


Fig. 12. Performance of the proposed method with different transformer models on the Santa Barbara dataset.

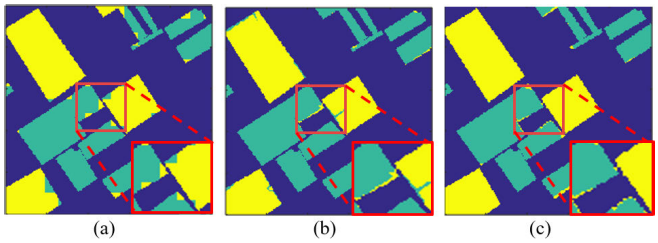


Fig. 13. Subjective visual comparison of the proposed method with different transformer models on the Santa Barbara dataset. (a) Transformer. (b) Graph-Transformer. (c) LIEG.

D. Ablation Studies

1) Performance With Different LIEG Block Numbers: We compare the performance of the proposed method using different LIEG block numbers on all the four datasets to determine the optimal number of LIEG blocks. Experimental results shown in Fig. 11 reveal that when the number of LIEG blocks reaches 3, the proposed method performs best on the four datasets. When the number of LIEG blocks is larger than 3, the performance of the proposed method gradually decreases, as the number of blocks increases. As the number of LIEG blocks increases from 1 to 3, the depth of the network gradually increases, and deeper local–global feature information can be extracted, so that the performance of the network gradually improves. However, with the further increase in the number of LIEG blocks, the receptive field of the convolution layer in LIEG keeps getting larger, resulting in a gradual weakening of the local information enhancement effect in the LIEG block. Moreover, the increase in the number of LIEG blocks leads to the increase in parameters, which makes the network easy to overfit. The number of LIEG blocks is set to 3 in the following experiments, which is appropriate and effective on all datasets.

2) Effect of LIEG: To verify the effect of the LIEG block, we tested three cases of the proposed method on the Santa Barbara dataset. The first case is the proposed D-LIEG network, which is composed of multilayer LIEG blocks. The second and third cases are the proposed method composed of traditional patch-based transformer blocks and graph-transformer blocks, respectively. Traditional transformer block splits the input image into some regular patches. Their contrastive results on the Santa Barbara dataset with training sample size of 1% are given as "transformer," "graph-transformer," and "LIEG" in Figs. 12 and 13. The proposed graph-transformer performs better than the traditional transformer, which means that the performance is improved by considering the spatial–spectral correlation between pixels. From Fig. 13, it can be observed that the traditional transformer produces many correctly classified pixels. The graph-transformer obtains better results with less misclassified pixels, but the boundary of the changed areas

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

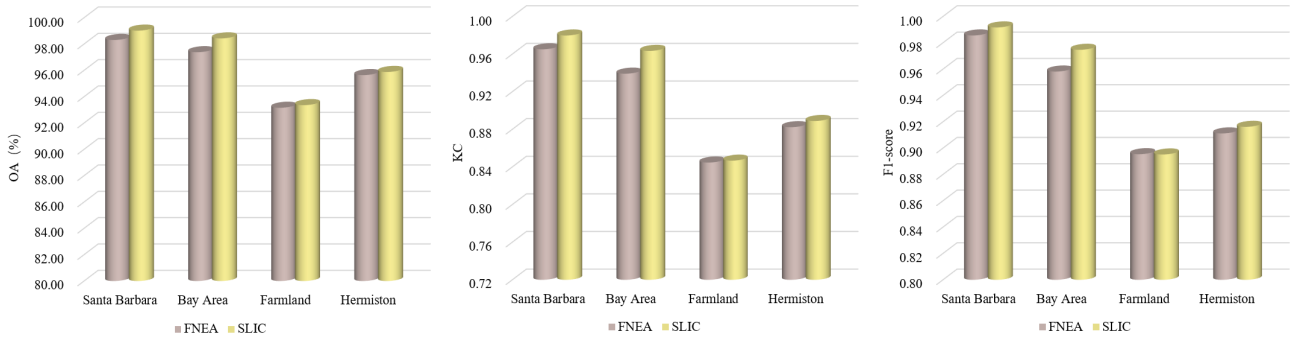


Fig. 14. Performance of the proposed method with different segmentation algorithms on four datasets.

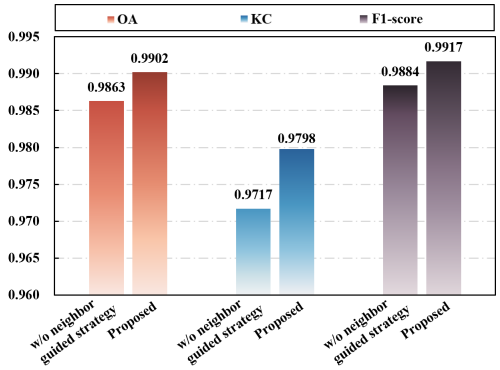


Fig. 15. Performance comparison of the proposed method with and without neighbor-guided strategy in graph-transformer on the Santa Barbara dataset.

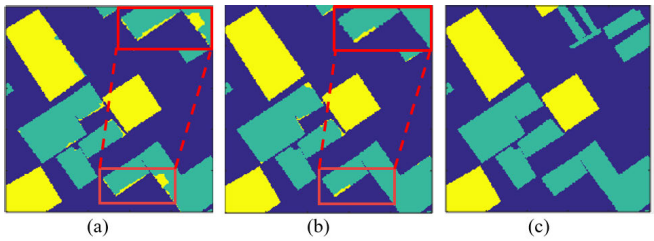


Fig. 16. Subjective visual comparison of the proposed method with and without neighbor-guided strategy in graph-transformer on the Santa Barbara dataset. (a) W/o neighbor-guided strategy. (b) Proposed. (c) Ground Truth.

is failed to be detected. The LIEG block, which combines the locality of convolution and the global representation capability of graph-transformer, achieves the best results, demonstrating the effectiveness of the LIEG block.

3) Effect of Segmentation Algorithm: Due to its efficiency and simple implementation, the SLIC algorithm is used to partition the images into superpixels in this article. To prove the robustness of the proposed method to the segmentation algorithm, we replace the SLIC with fractal net evolution approach (FNEA) [48] in the proposed method and conduct ablation experiments. FNEA is a widely used segmentation algorithm that has been proven to be effective in remote sensing image analysis. The OA, KC, and F1-score values of the results are shown in Fig. 14. As can be observed, the proposed method still achieves good performance when using the FNEA algorithm for segmentation. The results show that the proposed method is not sensitive to segmentation algorithm, demonstrating the robustness of the proposed method. Considering the subtle advantages of the SLIC algorithm, we adopt SLIC as the segmentation algorithm in the proposed method to obtain the nodes of graph.

4) Effect of Neighbor-Guided Strategy: We test two variants of the proposed method on the Santa Barbara dataset to investigate the effectiveness of the neighbor-guided strategy integrated in the attention operation. The model w/o neighbor-guided strategy is a variant of the proposed methods that removes the neighbor-guided strategy of the attention operation in the graph-transformer. The quantitative comparison results of different models are shown in Fig. 15. It can be observed that the proposed method with neighbor-guided attention operation achieves the better performance than the model w/o neighbor-guided strategy in terms of both OA, KC, and F1-score. There exist missed detection and false detection phenomenon in the visual results shown in Fig. 16(a). The change map of the proposed method shows higher consistency with the ground-truth map. Experimental results show that the proposed neighbor-guided attention operation considers the neighbor nodes with the highest similarity, which effectively highlights the useful spatial–spectral information for HSI-CD.

V. Conclusion

In this article, we propose a D-LIEG architecture for HSI-CD task. The transformer that can model global features is introduced to solve the HSI-CD problem, and the graph-transformer is innovatively designed to maintain the spatial–spectral correlation between graph nodes with low computational cost. In order to alleviate the loss of local information of graph-transformer, we further inject the local information obtained by the convolution into the graph-transformer. The dual-branch structure, where each branch is composed of multiple LIEG blocks, is used to extract sufficient features of multitemporal HSIs and send them to a classifier to realize the discrimination of changed objects. Extensive experiments have indicated that our method has achieved superior performance in both quantitative and qualitative results.

Although the proposed method shows good performance with limited training samples, the model training process still depends on manually annotated labeled samples. In the future, we will investigate a novel graph-based strategy to generate sufficient labeled data, which further reduces the dependence

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

of the proposed method on labeled samples, thus extending our research to unsupervised CD.

REFERENCES

[1] S. Liu, M. Chi, Y. Zou, A. Samat, J. A. Benediktsson, and A. Plaza, "Oil spill detection via multitemporal optical remote sensing images: A change detection perspective," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 324–328, Mar. 2017.

[2] L. Bruzzone and F. Bovolo, "A novel framework for the design of change-detection systems for very-high-resolution remote sensing images," *Proc. IEEE*, vol. 101, no. 3, pp. 609–630, Mar. 2013.

[3] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[4] W. Dong, S. Hou, S. Xiao, J. Qu, Q. Du, and Y. Li, "Generative dual-adversarial network with spectral fidelity and spatial enhancement for hyperspectral pansharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7303–7317, Dec. 2022.

[5] W. Dong, T. Zhang, J. Qu, S. Xiao, J. Liang, and Y. Li, "Laplacian pyramid dense network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507113.

[6] Y.-J. Deng, H.-C. Li, X. Song, Y.-J. Sun, X.-R. Zhang, and Q. Du, "Patch tensor-based multigraph embedding framework for dimensionality reduction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1630–1643, Mar. 2020.

[7] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.

[8] A. Asokan and J. Anitha, "Change detection techniques for remote sensing applications: A survey," *Earth Sci. Informat.*, vol. 12, no. 2, pp. 143–160, Jun. 2019.

[9] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[10] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[11] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.

[12] S. T. Seydi and M. Hasanlou, "A new land-cover match-based change detection for hyperspectral imagery," *Eur. J. Remote Sens.*, vol. 50, no. 1, pp. 517–533, Jan. 2017.

[13] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[14] V. Ortizrivera, M. Vélezreyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, May 2006, Art. no. 623312.

[15] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[16] B. Wang, S.-K. Choi, Y.-K. Han, S.-K. Lee, and J.-W. Choi, "Application of IR-MAD using synthetically fused images for change detection in hyperspectral data," *Remote Sens. Lett.*, vol. 6, no. 8, pp. 578–586, Aug. 2015.

[17] M. Hearst, "Support vector machines," *IEEE Intell. Syst. Their Appl.*, vol. 13, no. 4, pp. 18–21, Jul./Aug. 1998.

[18] R. A. A. Raja, V. Anand, A. S. Kumar, S. Maithani, and V. A. Kumar, "Wavelet based post classification change detection technique for urban growth monitoring," *J. Indian Soc. Remote Sens.*, vol. 41, no. 1, pp. 35–43, Mar. 2013.

[19] P. Lv, Y. Zhong, J. Zhao, and L. Zhang, "Unsupervised change detection based on hybrid conditional random field model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4002–4015, Jul. 2018.

[20] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[21] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[22] Y.-J. Deng, Y. Zhou, W.-Y. Wang, X.-H. Zhu, and H.-C. Li, "Robust patch tensor-based multigraph embedding for dimensionality reduction of hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1149–1152.

[23] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[24] J. Qu, S. Hou, W. Dong, S. Xiao, Q. Du, and Y. Li, "A dual-branch detail extraction network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518413.

[25] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Apr. 20, 2023, doi: 10.1109/tcsvt.2023.3268757.

[26] Y. Lin, S. Li, L. Fang, and P. Ghamisi, "Multispectral change detection with bilinear convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1757–1761, Oct. 2020.

[27] F. Gao, X. Liu, J. Dong, G. Zhong, and M. Jian, "Change detection in SAR images based on deep semi-NMF and SVD networks," *Remote Sens.*, vol. 9, no. 5, p. 435, May 2017.

[28] T. Zhan et al., "TDSSC: A three-directions spectral–spatial convolution neural network for hyperspectral image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 377–388, 2021.

[29] W. Dong et al., "Abundance matrix correlation analysis network based on hierarchical multihead self-cross-hybrid attention for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501513.

[30] H. Lyu, H. Lu, and L. Mou, "Learning a transferable change rule from a recurrent neural network for land cover change detection," *Remote Sens.*, vol. 8, no. 6, p. 506, Jun. 2016.

[31] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[32] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[33] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[34] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.

[35] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[36] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* in Lecture Notes in Computer Science, vol. 12346, A. Vedaldi, H. Bischof, T. Brox, and J. Frahm, Eds. Glasgow, U.K.: Springer, Aug. 2020, pp. 213–229.

[37] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[38] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.

[39] Y. Jiang, S. Chang, and Z. Wang, "TransGan: Two pure transformers can make one strong GAN, and that can scale up," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 14745–14758.

[40] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, Jun. 2020, pp. 5790–5799.

[41] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.

[42] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.
[43] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.
[44] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multiscale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.
[45] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.
[46] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
[47] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.
[48] M. Baatz and A. Schape, "Multiresolution segmentation: An optimization approach for high quality multi-scale image segmentation," in *Proc. Beiträge Zum AGIT-Symp.*, 2000, pp. 1–6.



Wenqian Dong (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She has been a Visiting Scholar with Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has authored or coauthored more than 20 papers in refereed journals and conferences, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Remote Sensing*. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.



Yufei Yang (Student Member, IEEE) received the B.E. degree in communication engineering from Hunan Normal University, Changsha, China, in 2019. She is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, China.

Her research interests include hyperspectral image fusion, change detection, and deep learning.



Jiahui Qu (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was an exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has authored or coauthored over 20 papers in academic journals and conferences, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Remote Sensing*. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



Song Xiao received the M.S. degree in communication and information system and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

From 2006 to 2007, she was with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA, as a Post-Doctoral Researcher. She is currently a Professor with the Beijing Electronic Science and Technology Institute, Beijing, China, and also a Professor with the School of Telecommunications Engineering, Xidian University. She has authored over 80 international journal and conference papers. Her research interests include image processing, image compression and coding, joint source channel coding, multimedia transmission systems over wired/wireless network, and compressed sensing.

Dr. Xiao is a Council Member of the Shaanxi Society of Image and Graphics and is a member of the IEEE Multimedia Communication Technology Committee and the IEEE Signal Processing Society. She is the Secretary-General of the Image Application in Military and Civilian Integration (IAMCI) Professional Committee of China Society of Image and Graphics.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is currently a Professor and also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:04:44 UTC from IEEE Xplore. Restrictions apply.